

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes

Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Sowmiya. D (192511063) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sowmiya. D

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand
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1. Intelligent Agents in a Smart Home Automation System

A **smart home automation system** uses intelligent agents to monitor the environment, make decisions, and perform actions automatically. The system receives inputs such as **motion detection, temperature changes, time of day, and user behavior** to manage lighting, heating/cooling, and security. Different types of intelligent agents can be applied depending on the complexity of the task.

i) Simple Reflex Agent

A **Simple Reflex Agent** makes decisions based only on the current percept (current input) using predefined **condition-action rules**. It does not consider past events or future consequences.

How it functions in the smart home:

- Turns on lights when motion is detected in a room.
- Switches off lights when no motion is detected for a specified time.
- Activates security alarms immediately when unauthorized motion is detected.

Advantages:

- Fast response to real-time events.
- Easy to implement.
- Requires minimal computational resources.

Limitations:

- Cannot learn user habits.
- Cannot remember previous states.
- May waste energy if decisions rely only on immediate inputs.

Example:

- If motion is detected after sunset → **Turn on hallway lights.**

ii) Model-Based Reflex Agent

A **Model-Based Reflex Agent** maintains an **internal model** of the environment. It remembers previous states and uses that information to make better decisions.

How it functions in the smart home:

- Remembers whether a room is occupied.
- Tracks indoor temperature trends before adjusting the thermostat.
- Recognizes that windows are open before starting air conditioning.
- Monitors whether doors were previously locked.

Advantages:

- More accurate than a simple reflex agent.
- Handles partially observable environments.
- Reduces unnecessary actions.

Limitations:

- Does not optimize for long-term goals.
- Requires memory and additional processing.

Example:

- If no motion has been detected for 30 minutes and the room was previously occupied
→ Turn off lights and reduce air conditioning.

iii) Goal-Based Agent

A **Goal-Based Agent** chooses actions that help achieve specific objectives. It evaluates different possible actions before deciding.

Goals in a smart home may include:

- Maintain comfortable room temperature.
- Minimize electricity consumption.
- Keep the house secure.
- Ensure user convenience.

How it functions:

- Determines the best heating schedule to reach the desired temperature before the user arrives home.
- Locks all doors automatically before bedtime.
- Selects the most efficient lighting schedule.

Advantages:

- Makes intelligent decisions based on objectives.
- Can plan future actions.
- More flexible than reflex agents.

Limitations:

- More computationally expensive.
- Requires planning algorithms.

Example:

- Goal: Maintain indoor temperature at **22°C** while minimizing energy usage.
- The agent delays air conditioning until necessary instead of running continuously.

4. Utility-Based Agent

A **Utility-Based Agent** selects actions that maximize overall satisfaction (utility). It evaluates different possible outcomes and chooses the one with the highest benefit by balancing multiple objectives.

How it functions in the smart home:

- Balances comfort, energy efficiency, and security simultaneously.
- Learns that the user prefers a cooler bedroom at night.
- Decides whether opening windows is more energy-efficient than using air conditioning.
- Adjusts lighting brightness according to user preferences and available daylight.

Advantages:

- Makes optimal decisions under uncertainty.
- Balances conflicting objectives.
- Can incorporate user preferences and learning.

Limitations:

- More complex to design.
- Requires utility functions and learning algorithms.

Example:

The agent evaluates:

- Comfort level
- Energy cost
- Security risk

It chooses the action that provides the **highest overall utility**, such as dimming lights while maintaining comfort and reducing electricity use.

Most Effective Approach: Hybrid Intelligent Agent

A **hybrid approach** combining **Model-Based, Goal-Based, and Utility-Based agents** is the most effective solution for a smart home automation system.

Why a Hybrid Approach?

- **Simple Reflex component:** Provides immediate responses to real-time events (e.g., turning on lights when motion is detected).
- **Model-Based component:** Maintains knowledge of occupancy, temperature history, and device status.
- **Goal-Based component:** Plans actions to achieve objectives such as maintaining comfort, reducing energy consumption, and improving security.

- **Utility-Based component:** Balances competing factors such as user comfort, electricity costs, environmental conditions, and safety while learning user preferences over time.

Benefits of the Hybrid Approach

- **Comfort:** Learns user routines and automatically adjusts lighting and temperature based on preferences.
- **Energy Efficiency:** Minimizes unnecessary power usage by optimizing heating, cooling, and lighting schedules.
- **Security:** Continuously monitors sensors, detects unusual activities, and automatically locks doors or triggers alarms when required.
- **Adaptability:** Improves performance over time by learning from user behavior and environmental changes.

Conclusion

While each intelligent agent type has specific strengths, **no single agent is sufficient** for a modern smart home. A **hybrid system** that combines **Simple Reflex, Model-Based, Goal-Based, and Utility-Based agents** offers the best performance. The reflex component ensures immediate responses, the model-based component maintains environmental awareness, the goal-based component achieves long-term objectives, and the utility-based component optimizes decisions by balancing **comfort, energy efficiency, and security**. This integrated approach creates a smart home that is responsive, adaptive, efficient, and capable of learning user preferences over time.

Robot Navigation in an Unknown Maze Using Intelligent Agents and Search Strategies

A robot placed in an **unknown maze** must explore its surroundings and find a path to the exit without prior knowledge of the maze layout. To achieve this, the robot uses **search algorithms** to explore possible paths and **intelligent agents** to decide which actions to take based on its perceptions and objectives.

1. Uniform Cost Search (UCS)

Uniform Cost Search (UCS) is an uninformed search algorithm that expands the node with the **lowest cumulative path cost** first. It is suitable when different paths have different movement costs.

How UCS Helps in Maze Navigation

- The robot starts from the initial position.
- It calculates the cumulative cost of each possible path.
- The path with the lowest total cost is explored first.
- The process continues until the exit is found.

Example:

If moving through one corridor costs 2 units and another costs 5 units, UCS chooses the lower-cost path first.

Advantages

- Finds the **optimal (least-cost)** path.
- Works well when movement costs vary.
- Complete if path costs are positive.

Disadvantages

- High memory usage because it stores all generated nodes.
- Can be slow in large mazes.
- Explores many unnecessary nodes before reaching the goal.

Complexity

- **Time Complexity:** $(O(b^{\lfloor 1 + \lfloor C^* / \epsilon \rfloor \rfloor}))$ (can be exponential)
- **Space Complexity:** $(O(b^{\lfloor 1 + \lfloor C^* / \epsilon \rfloor \rfloor}))$

Where:

- **b** = branching factor
- **C*** = cost of the optimal solution
- **ε** = minimum step cost

2. Iterative Deepening Search (IDS)

Iterative Deepening Search (IDS) combines the advantages of **Depth-First Search (DFS)** and **Breadth-First Search (BFS)**. It repeatedly performs depth-limited searches with increasing depth limits until the goal is found.

How IDS Helps in Maze Navigation

- The robot first searches up to depth 1.
- If the exit is not found, it searches up to depth 2.
- The depth limit increases until the exit is reached.

Advantages

- Requires very little memory.

- Complete for finite search spaces.
- Finds the shallowest solution when all step costs are equal.

Disadvantages

- Repeats exploration of upper-level nodes.
- Not optimal when movement costs differ.
- May take longer due to repeated searches.

Complexity

- **Time Complexity:** ($O(b^d)$)
- **Space Complexity:** ($O(bd)$)

Where:

- **b** = branching factor
- **d** = depth of the solution

Comparison of UCS and IDS

Feature	Uniform Cost Search (UCS)	Iterative Deepening Search (IDS)
Search Strategy	Lowest path cost first	Increasing depth limits
Optimal	Yes (least-cost path)	Yes (only when step costs are equal)
Complete	Yes	Yes
Time Complexity	Exponential	$(O(b^d))$
Space Complexity	Exponential	$(O(bd))$
Best Use	Variable movement costs	Equal movement costs and limited memory

Intelligent Agents in the Maze Problem

Different intelligent agents make decisions in different ways while navigating the maze.

1. Simple Reflex Agent

- Makes decisions based only on the current perception.
- Example: If there is a wall ahead, turn left.

Limitation: Cannot remember previously visited locations, so it may repeatedly enter the same dead ends.

2. Model-Based Agent

- Maintains an internal map of explored areas.
- Remembers visited paths and obstacles.
- Avoids revisiting the same locations unnecessarily.

Benefit: Performs much better in unknown environments because it uses memory.

3. Goal-Based Agent

- Focuses on achieving the goal of reaching the maze exit.
- Evaluates possible paths and chooses actions that move closer to the exit.
- Works effectively with search algorithms such as UCS or IDS.

4. Utility-Based Agent

- Chooses actions that maximize overall utility.
- Considers multiple factors such as:
 - Shortest path
 - Lowest energy consumption
 - Minimum travel time
 - Safety

Instead of simply reaching the exit, it selects the most beneficial path according to these criteria.

Effect of Agent Design on Decision-Making

The design of the intelligent agent directly affects how efficiently the robot solves the maze.

- **Simple Reflex Agent:** Reacts quickly but lacks memory and often repeats mistakes.
- **Model-Based Agent:** Learns about the maze as it explores, improving navigation efficiency.
- **Goal-Based Agent:** Uses search algorithms to systematically find a route to the exit.
- **Utility-Based Agent:** Balances multiple objectives, such as minimizing energy use while still reaching the exit efficiently.

An agent with memory and planning capabilities generally performs much better than a purely reactive agent.

Most Effective Combination

The most effective solution is a **Model-Based Goal-Based Agent combined with Uniform Cost Search (UCS)**.

Justification

- **Model-Based Agent**
 - Builds an internal representation of the maze.
 - Avoids revisiting explored paths.
 - Handles unknown environments effectively.
- **Goal-Based Agent**
 - Keeps the robot focused on reaching the exit.
 - Uses search algorithms to plan efficient routes.
- **Uniform Cost Search**
 - Guarantees the least-cost path when movement costs differ.
 - Produces optimal solutions, which is valuable if different terrain or actions have different costs.

If **all movements have equal cost**, **Iterative Deepening Search (IDS)** is an excellent alternative because it requires much less memory while still guaranteeing a solution in finite mazes.

Conclusion

Both **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)** are effective uninformed search strategies for maze navigation. UCS is ideal for finding the **lowest-cost path** when movement costs vary, while IDS is preferred when **memory is limited** and all moves have equal cost. Among intelligent agents, **Model-Based** and **Goal-Based** agents are best suited for this task because they maintain knowledge of the explored maze and plan actions toward the exit. Therefore, a **Model-Based Goal-Based Agent with UCS** (or **IDS** when all step costs are equal) provides the most efficient and reliable solution for navigating an unknown maze.